Shake-down cases: general orbital roles

XPSoverlaps

1 Orbital roles

An initial state a is a single excitation $i \to k$ from the closed-shell reference. A shake-down goes to a lower-energy initial state b that differs from a by one moved electron; the core-ionised final state Φ^F has the valence occupation of b and one electron removed from the core orbital c. The roles are defined by occupation in the reference, not by a fixed HOMO/LUMO assignment:

Label	Role	Reference occupation	Physical orbital
c	core; doubly occupied initially, singly finally	occupied (core)	e.g. S 2p
i	pump hole: emptied by the pump	occupied	HOMO$-n$, $n \geq 0$
k	pump particle: filled by the pump	virtual	LUMO$+m$, $m \geq 0$
j (3e ES)	receives the shake-down electron	virtual, below k	LUMO$+l$, $l < m$
j (5e ES)	gives the shake-down electron; doubly occupied in a	occupied, above i	HOMO$-n'$, $n' < n$
j (5e GS)	doubly occupied throughout; kept in the closed-shell rule	occupied, above i	HOMO
d	HOMO when it is none of i, j (ES cases); doubly occupied throughout	occupied	HOMO

The spectator d carries no spin and is left out of the rules. Energy order: 3e cases $\varepsilon_i < \varepsilon_d < \varepsilon_j < \varepsilon_k$; 5e case $\varepsilon_i < \varepsilon_j < \varepsilon_d < \varepsilon_k$.

2 Which case applies

The case is set by *where the shake-down electron comes from*, read off by comparing b with a (both written as excitations from the reference):

b compared with $a = (i \to k)$	Shake-down electron	Case
b is the reference, i = HOMO	$k \to i$, from the virtual k into the hole	3e GS
b is the reference, i = HOMO$-n$	$k \to i$, with j = HOMO	5e GS
same hole i, lower particle	$k \to j$, from the virtual k into a lower virtual	3e ES
same particle k, different hole	$j \to i$, from a doubly occupied j into the hole	5e ES

Only one row can hold for a given pair, so an initial HOMO$-n$ →LUMO$+m$ state usually has several shake-downs, each with its own case. The 5e rules are needed only when an electron leaves an orbital that was *doubly occupied* in a; whenever the moving electron is the pumped one in k, the 3e rules apply.

3 Configurations

Figure 1 shows the occupations for each case; in short:

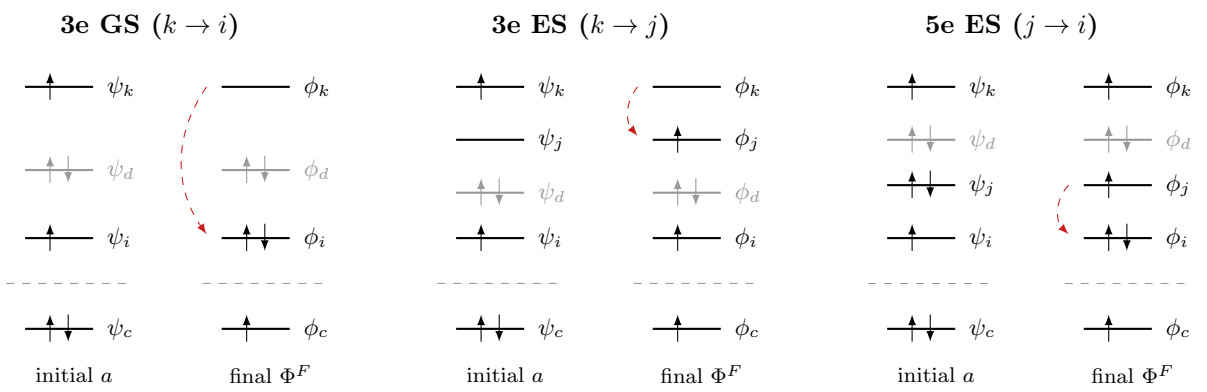


Figure 1: Occupations of the initial state a and the core-ionised final state Φ^F (valence as in b) for each case. Red: the shake-down electron. Grey: the HOMO spectator d, present only when it is none of i, j (i.e. $i = \text{HOMO}-n$ in 3e, $j \neq \text{HOMO}$ in 5e); otherwise d is absent and the diagram is unchanged. Dashed line: core–valence gap.

Case	Initial a	Final Φ^F	Lower initial state b
3e GS	$\psi_c^2\,\psi_i^1\,(\psi_d^2)\,\psi_k^1$	$\phi_c^1\,\phi_i^2\,(\phi_d^2)$	reference
3e ES	$\psi_c^2\,\psi_i^1\,(\psi_d^2)\,\psi_k^1$	$\phi_c^1\,\phi_i^1\,(\phi_d^2)\,\phi_j^1$	$i \to j$
5e ES	$\psi_c^2\,\psi_i^1\,\psi_j^2\,(\psi_d^2)\,\psi_k^1$	$\phi_c^1\,\phi_i^2\,\phi_j^1\,(\phi_d^2)\,\phi_k^1$	$j \to k$

4 Rules

$S_{ab} = |\langle\phi_a|\psi_b\rangle|$, where ϕ_a is the final-state orbital that the orbital mapping matches to initial orbital a. CVS: $S_{cc} = 1$, $S_{ca} = S_{ac} = 0$. Diagonal overlaps are set to $S_{ii} = S_{jj} = S_{kk} = 1$; the code reports any whose actual value is below 0.9.

	3e	5e
ES: Π (direct)	$S_{ii}S_{jk}$	M_1M_2
ES: Ω (exchange)	$S_{ik}S_{ji}$	M_3M_4
ES: singlet $\to D_1$	$\lvert\Pi+\Omega\rvert^2$	$\lvert\Pi+\Omega\rvert^2$
ES: triplet $\to D_2$, Q	$\frac{1}{3}, \frac{2}{3}\,\lvert\Pi-\Omega\rvert^2$	$\frac{1}{3}, \frac{2}{3}\,\lvert\Pi-\Omega\rvert^2$
GS: singlet $\to$ closed shell	$2\lvert S_{ii}S_{ik}\rvert^2$	$2\lvert M_1M_3\rvert^2$
GS: triplet	dark	dark

with the 2×2 minors of the valence overlap block

$$M_1 = S_{ii}S_{jj} - S_{ij}S_{ji}, \qquad M_2 = S_{ij}S_{kk} - S_{ik}S_{kj},$$
$$M_3 = S_{ij}S_{jk} - S_{ik}S_{jj}, \qquad M_4 = S_{ii}S_{kj} - S_{ij}S_{ki}.$$

3e GS is a HOMO pump; a GS shake-down from a HOMO$-n$ pump is 5e GS, with the HOMO as j (no spectator). The code keeps S_{ii}, S_{jj}, S_{kk} in the formulas it prints and sets them to 1.

5 Not computed

- a or b is a double excitation: no rule.
- The electron would move up ($\varepsilon_j > \varepsilon_k$ in 3e ES, or the donor below the hole in 5e ES).
- The order cannot be checked: with only HOMO and LUMO known, 3e ES needs $j = \text{LUMO}$ and 5e ES needs $j = \text{HOMO}$; other pairs are reported as "orbital order unknown".

6 Example

Bromothiophene, S 2p core hole; HOMO = mo:34, LUMO = mo:40.

$a \to b$	a	b	Electron	Case	i	j	k (d)
$2 \to 1$	mo:34→40	reference	40→34	3e GS	34	–	40
$5 \to 1$	mo:38→41	reference	41→38	5e GS	38	34	41
$5 \to 3$	mo:38→41	mo:38→40	41→40	3e ES	38	40	41 (34)
$5 \to 4$	mo:38→41	mo:34→41	34→38	5e ES	38	34	41
$6 \to 5$	mo:37→41	mo:38→41	38→37	5e ES	37	38	41 (34)

State 5 shows one pump with three shake-downs of different cases. $6 \to 5$ is a 5e ES with j = HOMO-1; it is computable once the order of mo:37 and mo:38 is known.